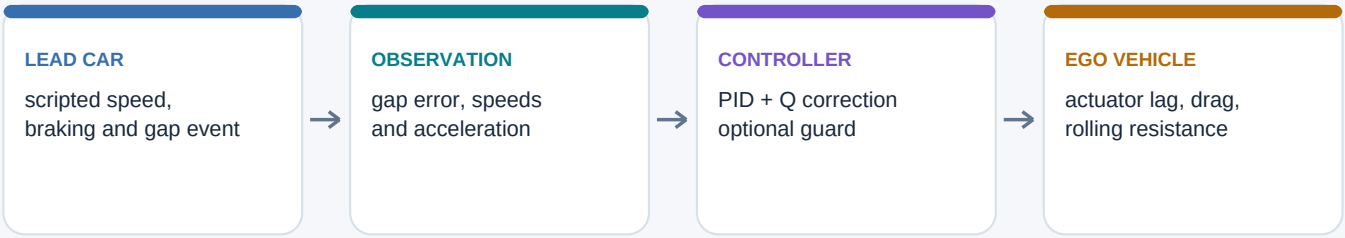
01 / PROJECT IN ONE MINUTE

A small learning layer inside a familiar ACC controller

This project asks whether a small tabular Q-learning correction, paired with a braking-distance action filter, can improve a classical adaptive cruise controller in a sudden-gap stress case.

THE CLOSED-LOOP SIMULATION



THE RESEARCH QUESTION

When the following gap drops suddenly, does filtering a tabular RL controller's proposed acceleration improve its response?

WHAT 50 MATCHED CUT-IN RUNS SHOWED

MINIMUM GAP

+4.35 m

Q + guard vs Q

SHORT-TTC SHARE

2.04% -> 0%

closing samples under 2 s

RIDE TRADE-OFF

0.957 -> 1.222

mean absolute jerk, m/s^3

Scope: The cut-in is an instantaneous longitudinal gap change. There is no lane-change geometry, perception model, or production-car head-to-head.

02 / CONTROLLER AND LEARNING

What each controller actually does

The learned policy does not steer or replace the whole controller. It adds one small acceleration correction to a transparent spacing controller; an optional filter can then alter that command.

CONTROLLER	ROLE IN THIS PROJECT
CTG-PID classical baseline	Targets a time gap: **desired distance = 5 m + 1.3 x ego speed**. A proportional gap term and relative-speed term produce a bounded acceleration command.
Q-learning tabular residual	Chooses one of seven residual accelerations and adds it to CTG-PID's proposed command. The learned object is a table of state-action values, not a large neural network.
Q + guard filtered action	Checks the proposed command against a simplified stopping-distance margin. If it fails, a bounded search selects the least-braking command that passes the check.
MPC predictive reference	Tests candidate commands using a short forward model of spacing, speed difference, acceleration and jerk. It is a simple model-based reference, not an OEM ACC system.

THE Q-LEARNING TABLE HAS THREE PARTS

STATE

5 binned signals:
gap error, closing speed, ego speed, ego acceleration, and lead acceleration.

ACTION

7 residual accelerations:
-1.2, -0.6, -0.25, 0, +0.25, +0.6, +1.2 m/s^2.

REWARD

Favors desired spacing and smooth motion; penalizes closing speed, weak stopping margin and collision.

Q-LEARNING UPDATE

$$Q(s,a) \leftarrow Q(s,a) + \alpha [\, r + \gamma \max Q(s') - Q(s,a) \,]$$

Training uses alpha = 0.12, gamma = 0.985, and epsilon-greedy exploration that decays from 1.0 to 0.05.

Training profiles are steady speed, a smooth speed wave, and moderate lead-car braking. The cut-in and emergency profiles are held out for evaluation.

WHY THIS STAYS LIGHTWEIGHT

The binned state space has 9 x 10 x 8 x 7 x 7 = 35,280 states. With 7 actions, the learned table stores 246,960 values - compact enough to inspect without a deep network or GPU.

03 / SIMULATION AND EXPERIMENT DESIGN

Learn on routine driving; test stressful events separately

The experiment compares methods under matched initial conditions. Every method gets the same event parameters for a given seed, so the comparison is paired rather than a set of unrelated runs.

FORMAL EVALUATION

50 seeds x 5 scenarios x 4 controllers = 1,000 episodes

Each episode: 40 seconds at 0.1-second steps. Controllers share each seed's start and event.

SCENARIO	WHAT HAPPENS	ROLE
Steady	Lead speed stays approximately constant.	Training + evaluation
Wave	Lead speed varies smoothly around its base speed.	Training + evaluation
Moderate brake	Lead vehicle slows and later recovers.	Training + evaluation
Cut-in	Gap drops instantly by 16-26 m; lead then brakes.	Held-out test
Emergency	Higher-speed case with stronger lead braking.	Held-out test

WHAT THE SIMULATOR INCLUDES

VEHICLE

1,500 kg point mass; acceleration command limited to -6 to +2.5 m/s^2.

ACTUATOR

First-order lag with a 0.35 s time constant.

ROAD LOAD

Aerodynamic drag and rolling resistance.

GAP RULE

5 m standstill spacing + 1.3 s time gap at ego speed.

METRICS TO READ

METRIC	PLAIN-LANGUAGE MEANING
Minimum gap	Smallest recorded bumper-to-bumper spacing in an episode.
TTC under 2 s	Share of closing samples where gap / closing speed is below 2 seconds.
Mean absolute jerk	Average size of acceleration change per second; a rough ride-smoothness proxy.
Gap RMSE	How closely measured gap follows the desired time-gap target.

04 / RESULTS AND INTERPRETATION

What happened in the 50 cut-in episodes

The cut-in family is a synthetic holdout: the Q table was trained on steady, wave, and moderate-braking profiles, not on cut-in episodes.

METHOD	MIN GAP (m)	TTC UNDER 2 s (%)	ABS JERK (m/s^3)	GAP RMSE (m)
Classical CTG-PID	8.58	0.40	0.256	5.98
Q-learning	7.00	2.04	0.957	6.96
Q + braking guard	11.35	0.00	1.222	5.40
Predictive MPC	13.11	0.00	0.630	3.76

READ THE TRADE-OFF

Q + guard vs Q: mean episode minimum gap rises by 4.35 m; closing-sample TTC-under-2 share changes from 2.04% to 0%.
Comfort cost: mean absolute jerk changes from 0.957 to 1.222 m/s^3.
Reference: MPC has the largest mean minimum gap (13.11 m) and the lowest cut-in gap RMSE (3.76 m).

WHAT YOU CANNOT CLAIM

No collisions were observed in these 50 runs, but this is not a risk estimate or safety proof. The guard is RSS-inspired, not a formal RSS implementation. CTG-PID is a classical simulated baseline; no production vehicle was tested under matched conditions.

A separate 2025 Honda Accord manual describes warnings for abrupt lead-vehicle deceleration and cut-ins. That is feature context only, not evidence that this controller outperforms a commercial ACC system.

REPORTING TIP

Report episode means and spread, not one representative seed. For a matched-method claim, use the per-seed deltas in **paired_differences.csv**. The displayed 95% intervals are descriptive normal approximations, not significance tests. Describe the TTC result as a share of closing samples under two seconds.

05 / RUN IT AND EXPLORE THE CODE

A practical route through the project

The simulator and tabular policy run on CPU. The formal batch previously used about 42 MB peak resident memory on this machine. MP4 rendering requires FFmpeg in addition to the Python packages.

1. INSTALL THE PROJECT PACKAGES

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
```

2. TRAIN THE Q TABLE

```
python src/train_q_learning.py --episodes 6000 --seed 17 --output models/q_agent.npz
```

3. EVALUATE MATCHED EPISODES

```
python src/run_experiments.py --seeds 50 --output outputs/formal_50seed
```

4. RENDER THE EXPLAINER

```
python src/make_explainer_video.py --output workspace/output/ACC_CutIn_Explainer.mp4
```

START HERE IN THE SOURCE TREE

FILE	PURPOSE
src/acc_sim.py	Vehicle model, scenarios, CTG-PID, guard, MPC, metrics.
src/train_q_learning.py	Exploration, reward and Q-table update.
src/run_experiments.py	Matched seeds, scenario summaries and trace export.
src/make_explainer_video.py	Plots the recorded cut-in trace and 50-seed results.
outputs/formal_50seed/	Reproducible metrics, manifest, traces and comparison clips.

KEEP LEARNING

Read README.md first; use the 19-page project guide for literature and method detail; use the focused presentation for the class narrative. Dataset notes explain why the public traces are context, not a matched cut-in benchmark.